

Appendix A - Geographe Marine Park Natural Values Report

South-west Marine Parks Network

Table of contents

Natural Values	1
Benthic ecosystem extent	2
Observations	2
Dominant habitat types	3
Habitat distribution and probability	4
Functional reef distribution and probability	5
Benchmarks of benthic natural values	6
Seagrass	6
Macroalgae	7
Sessile invertebrates	8
Benchmarks of demersal fish natural values	9
Survey effort	9
Species richness	10
Total abundance	12
Large Reef Fish Index*	14
Reef fish thermal index	16

Natural Values

Understanding of natural values

Benthic ecosystem extent

Observations

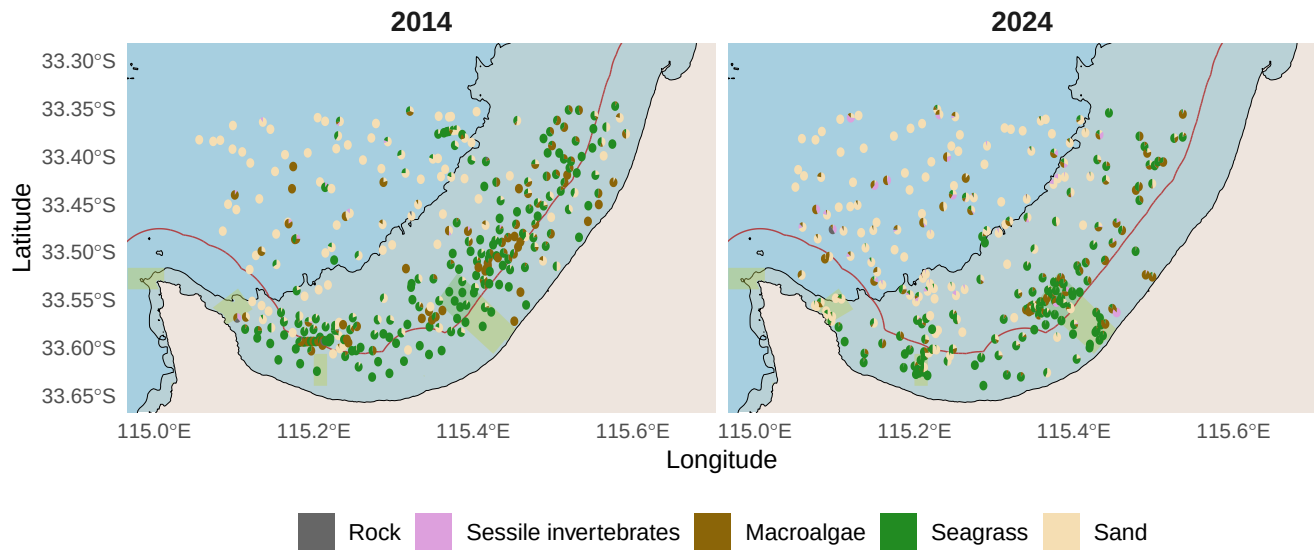


Figure 1: Habitat classes observed in stereo-BRUV imagery. Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

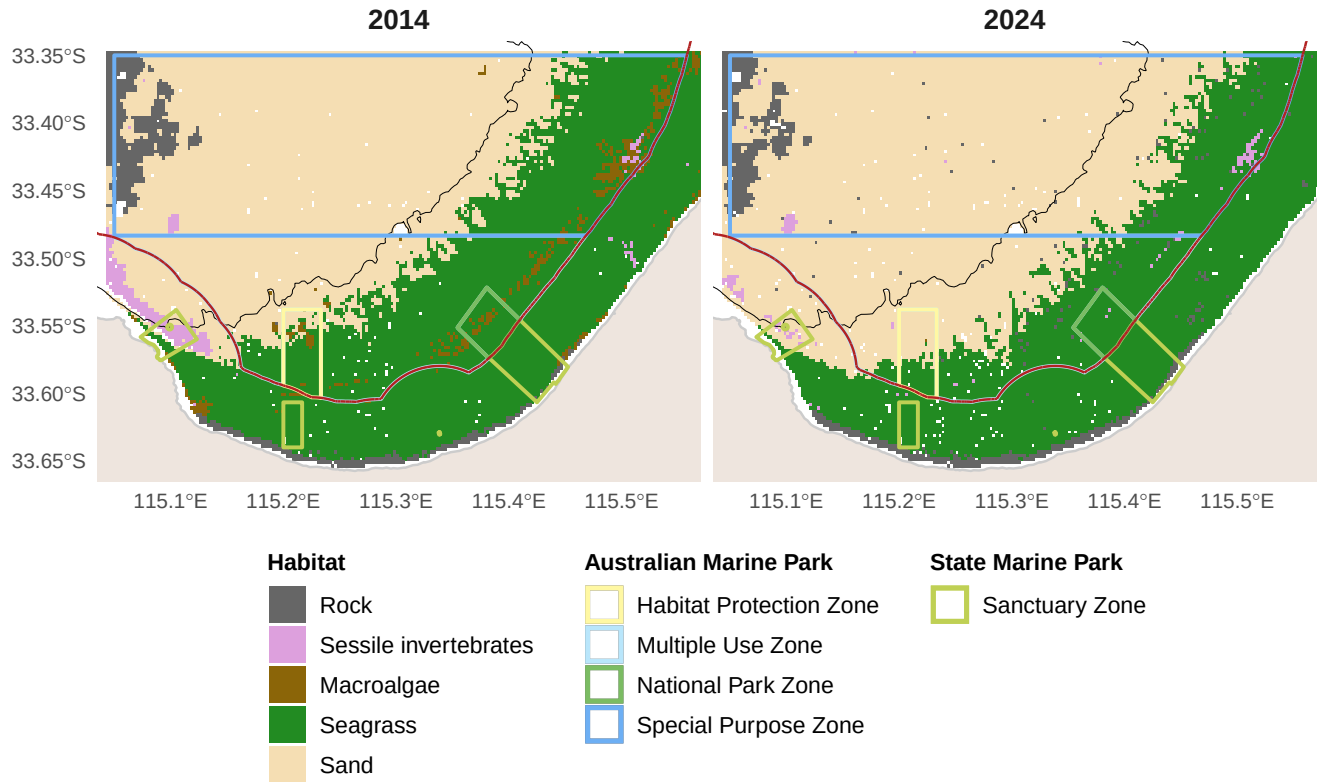


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones and State no-take sanctuary zones. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

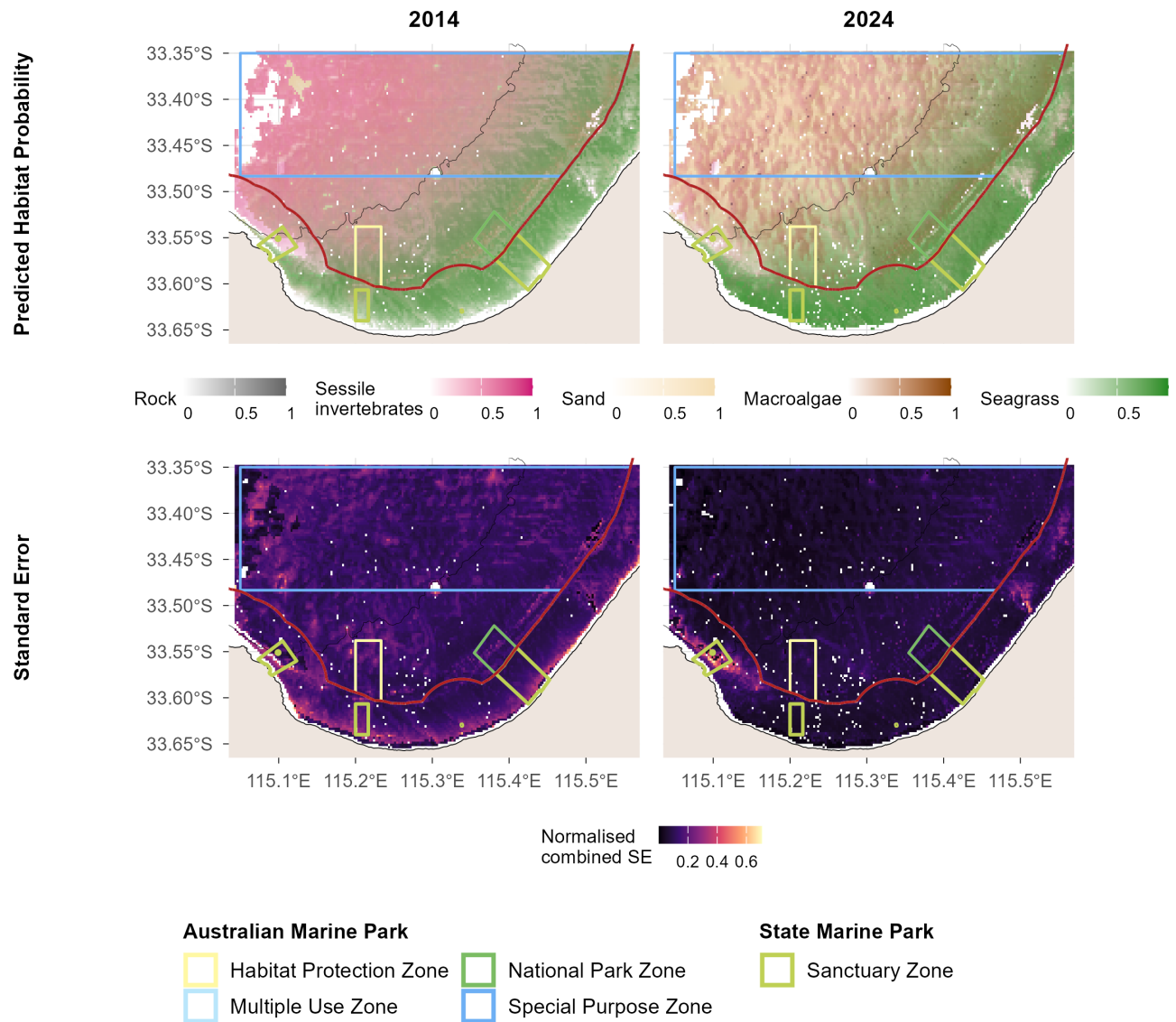


Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

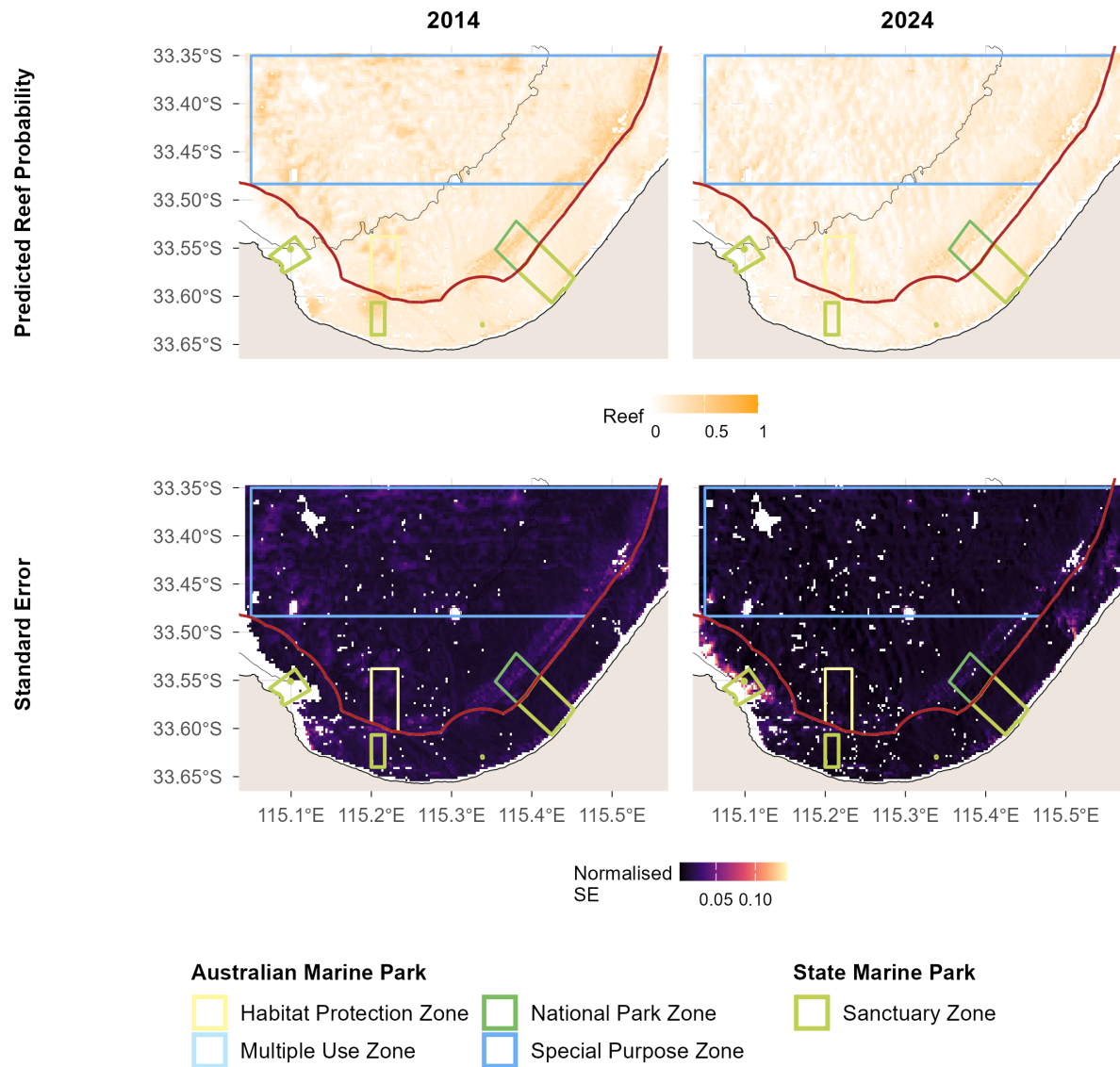


Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Seagrass

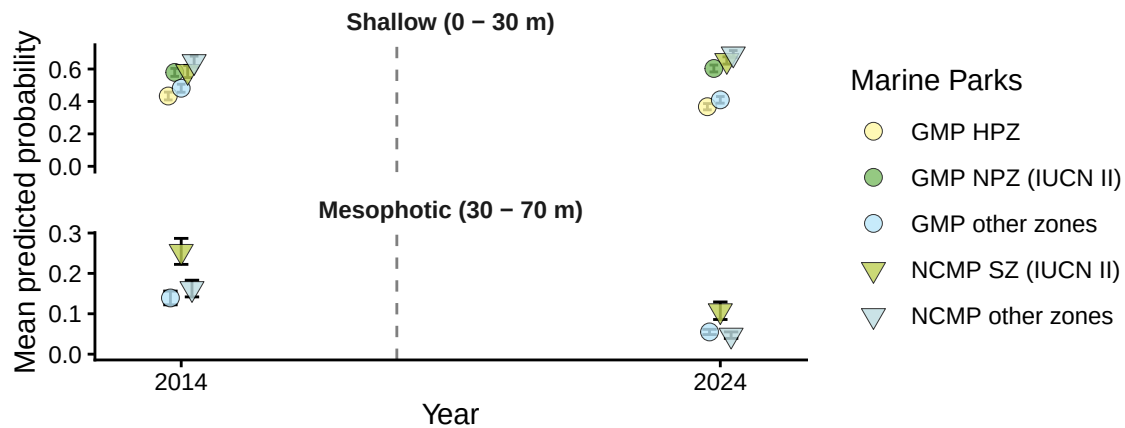


Figure 5.1: Seagrass: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

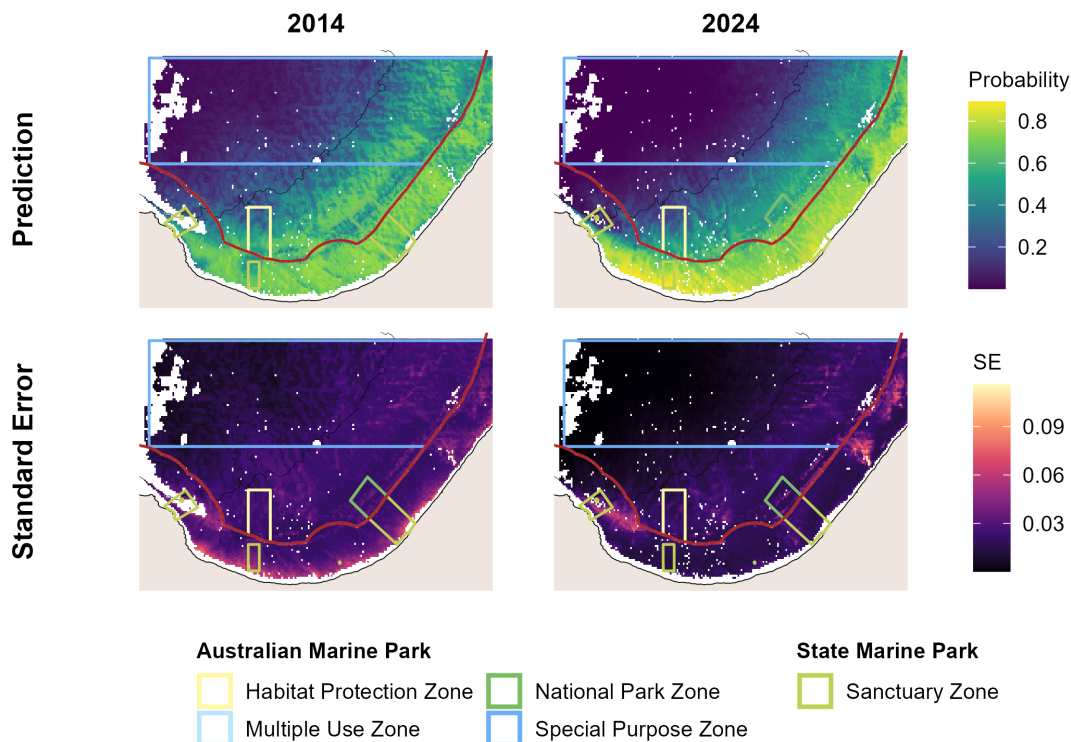


Figure 5.2: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

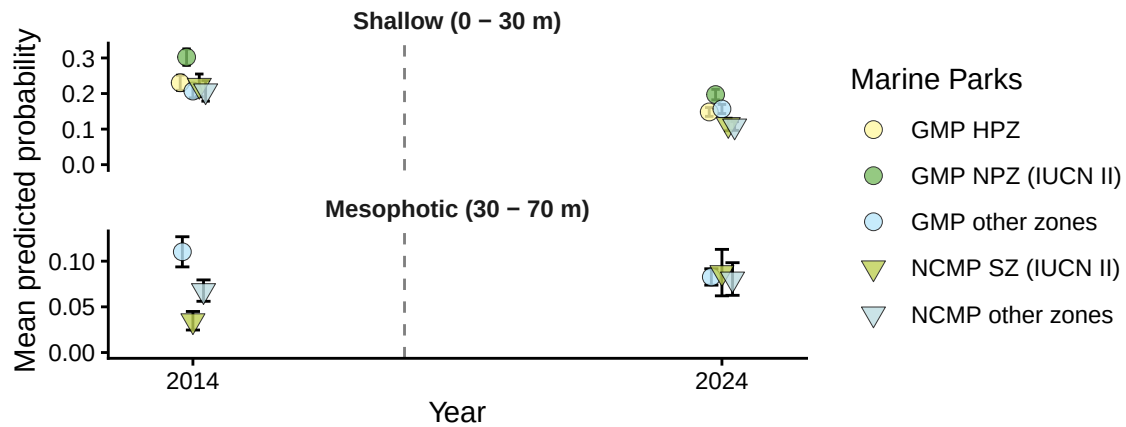


Figure 6.1: Macroalgae: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

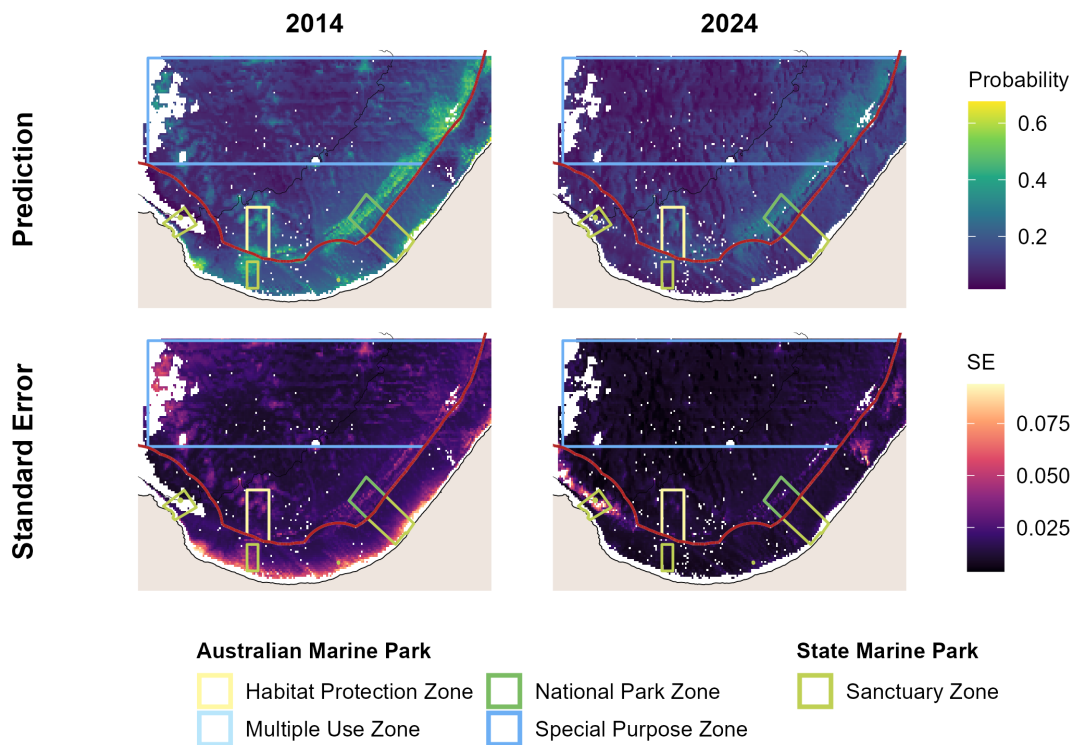


Figure 6.2: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

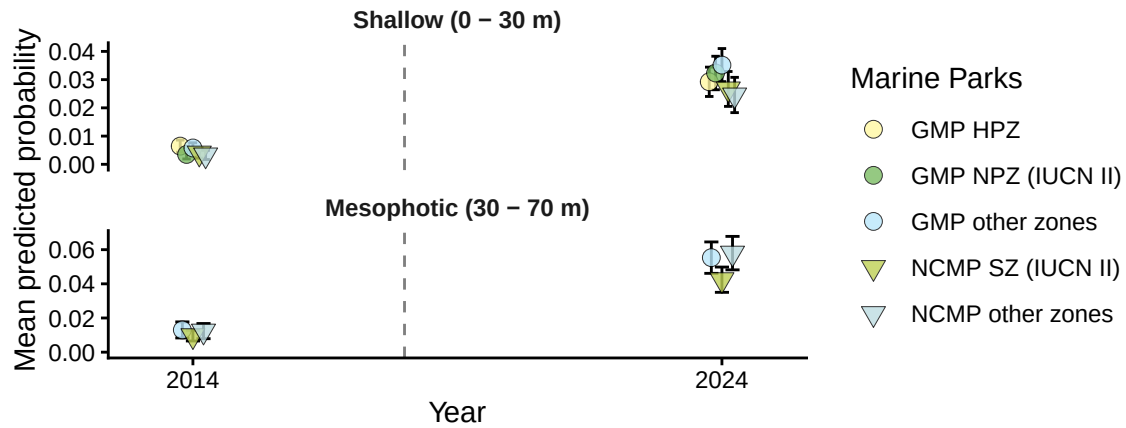


Figure 7.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

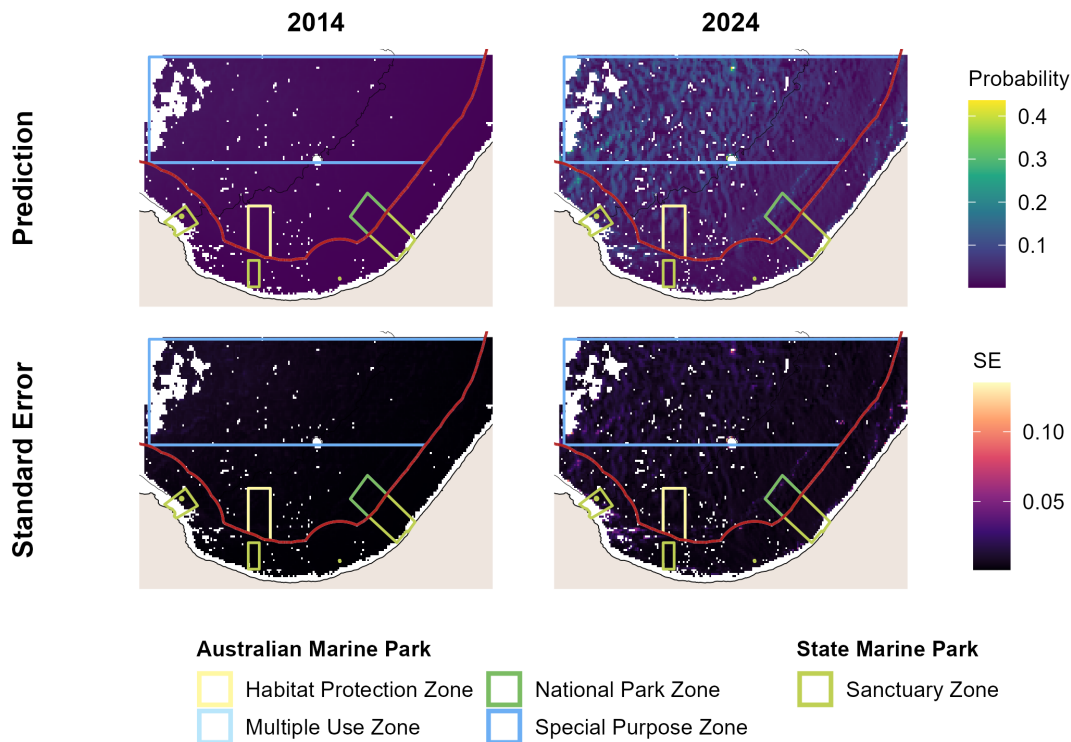
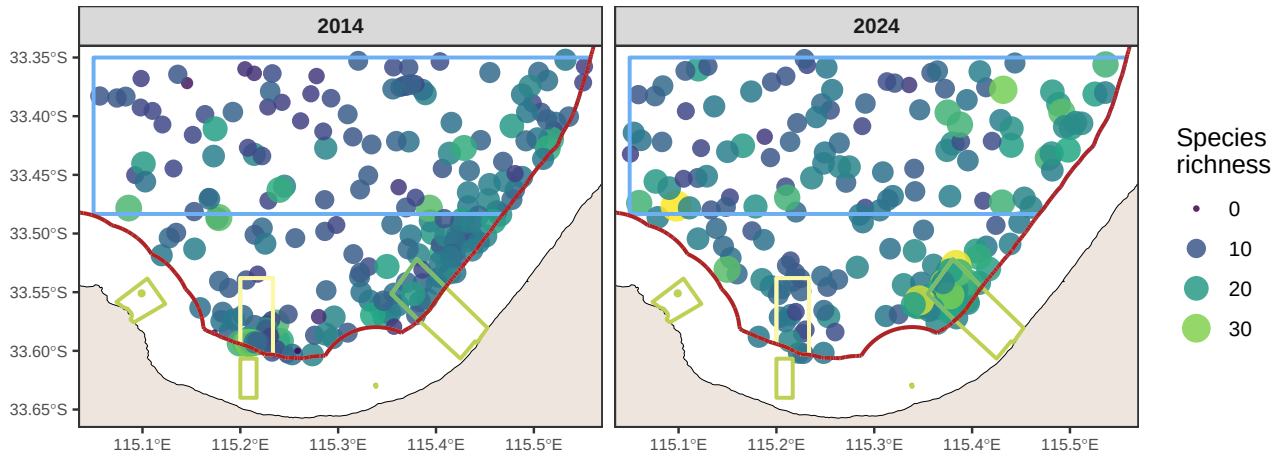


Figure 7.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

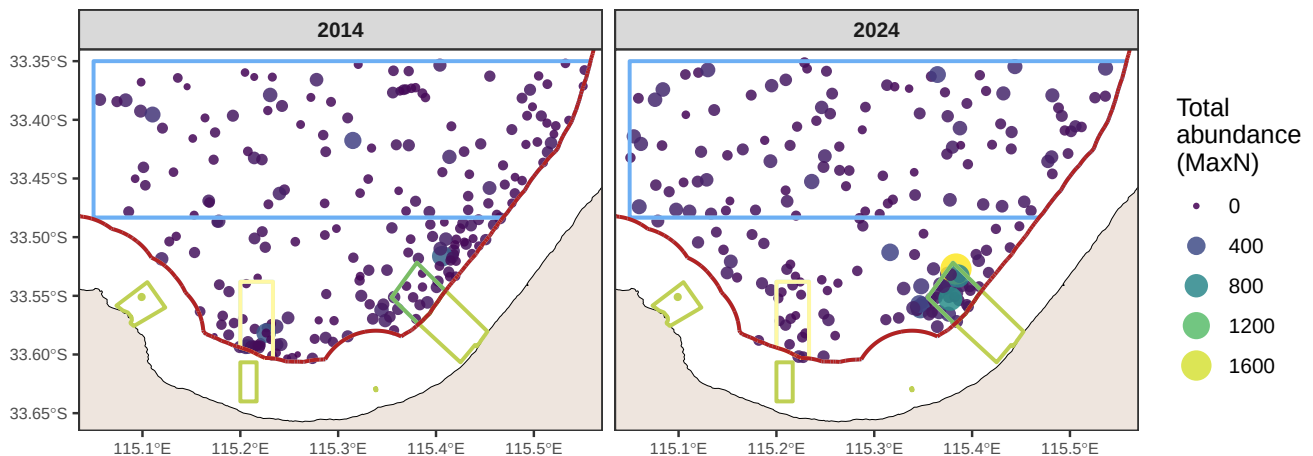
Benchmarks of demersal fish natural values

Survey effort

a)



b)



Australian Marine Park

- Habitat Protection Zone
- Multiple Use Zone

- National Park Zone
- Special Purpose Zone

State Marine Park

- Sanctuary Zone

Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

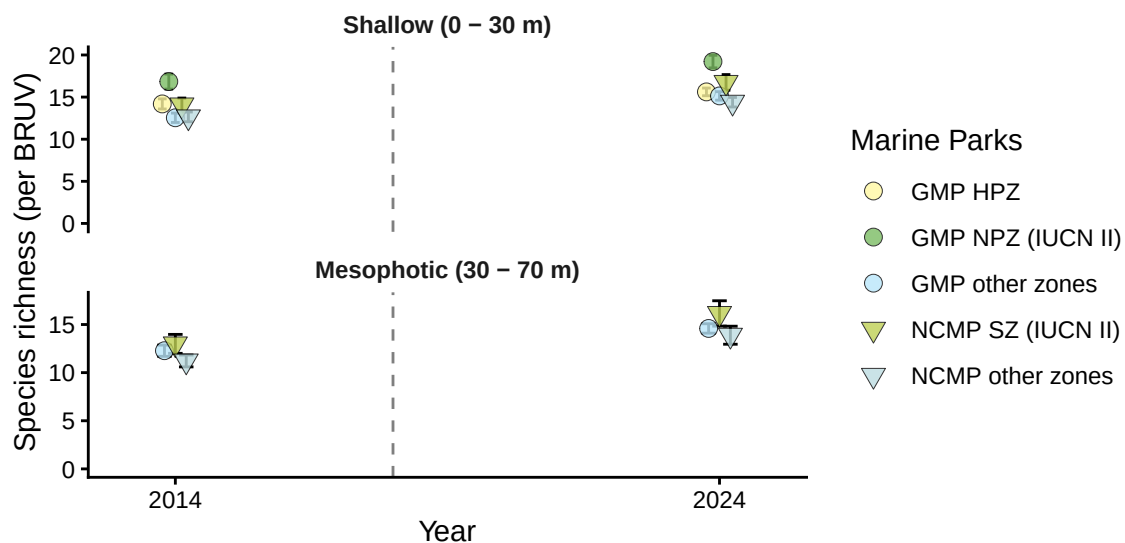


Figure 9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

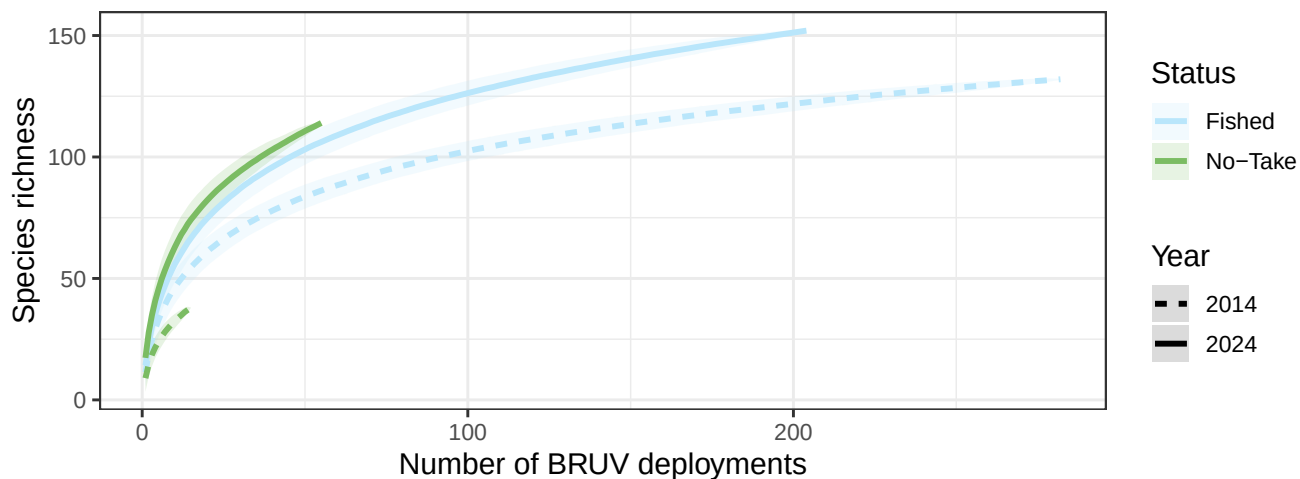


Figure 9.2: Demersal fish - Species richness: species accumulation curve by zone type from stereo-BRUV groundtruthing.

Species richness

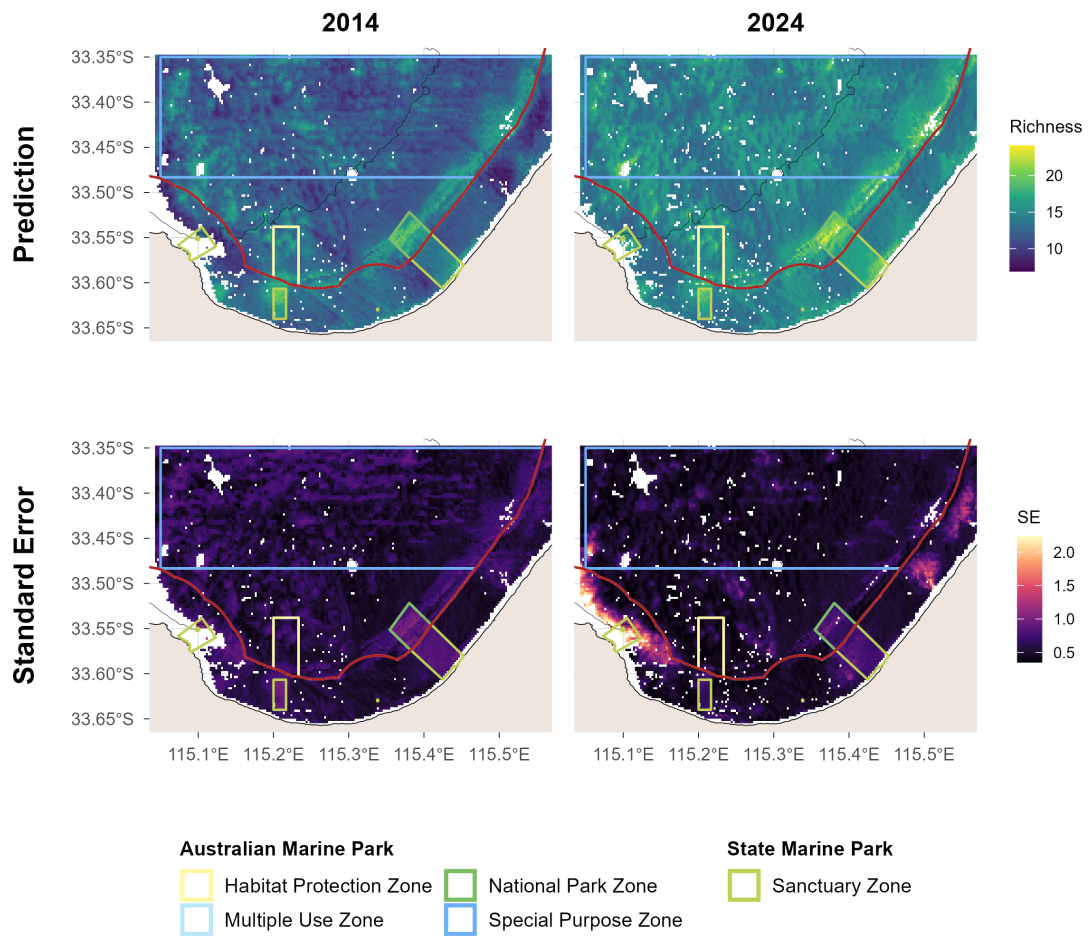


Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

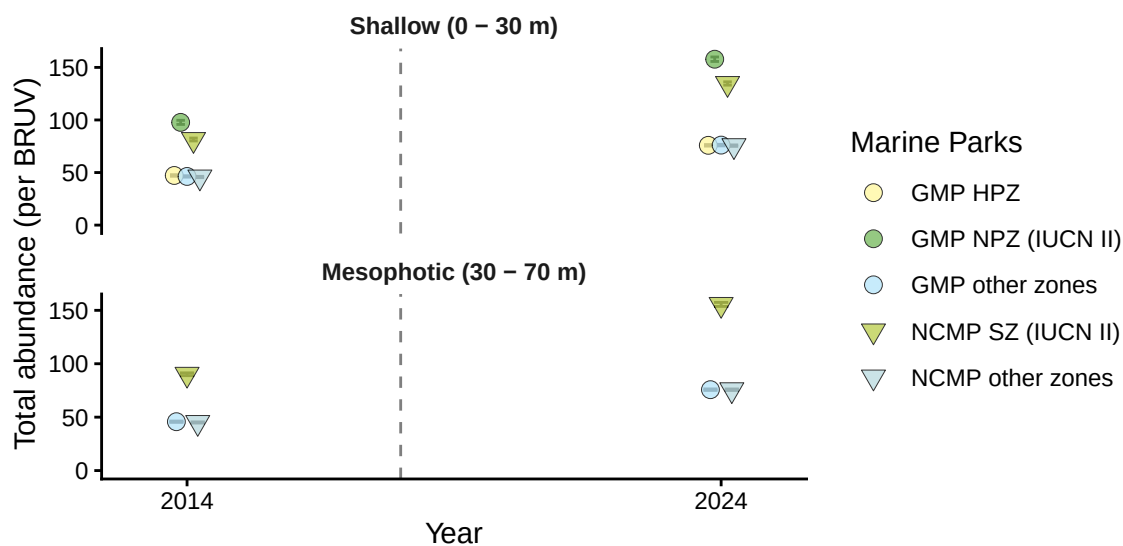


Figure 10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

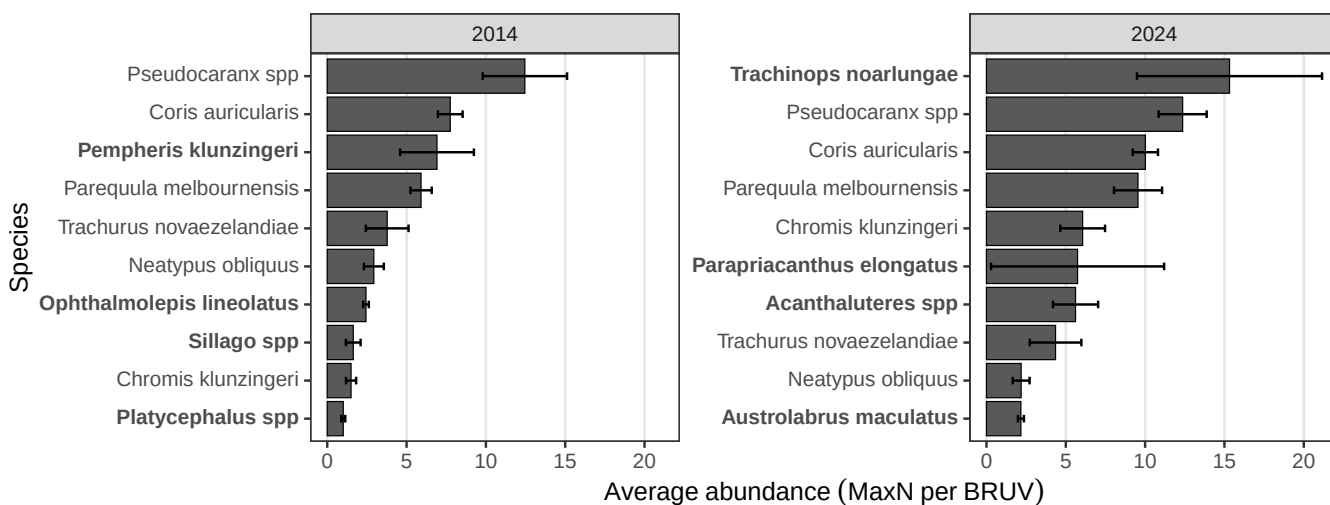


Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Total abundance

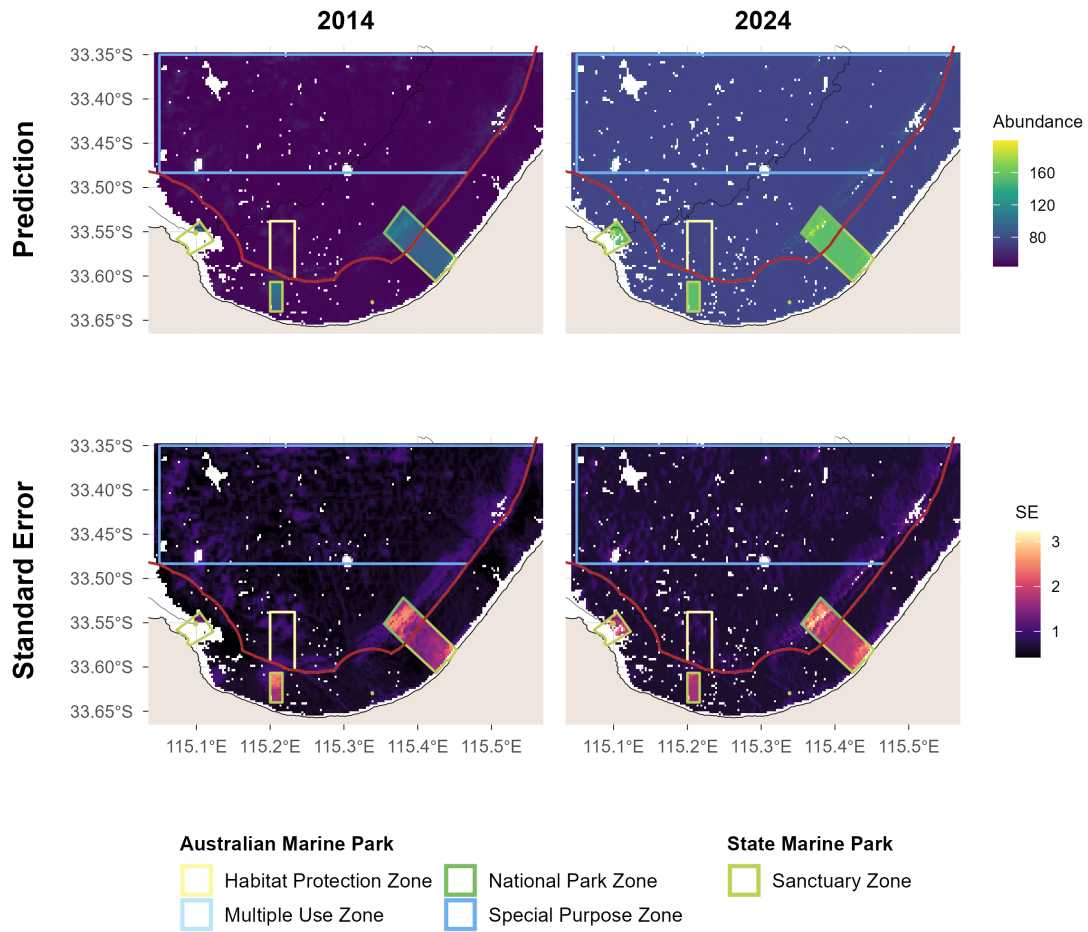


Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

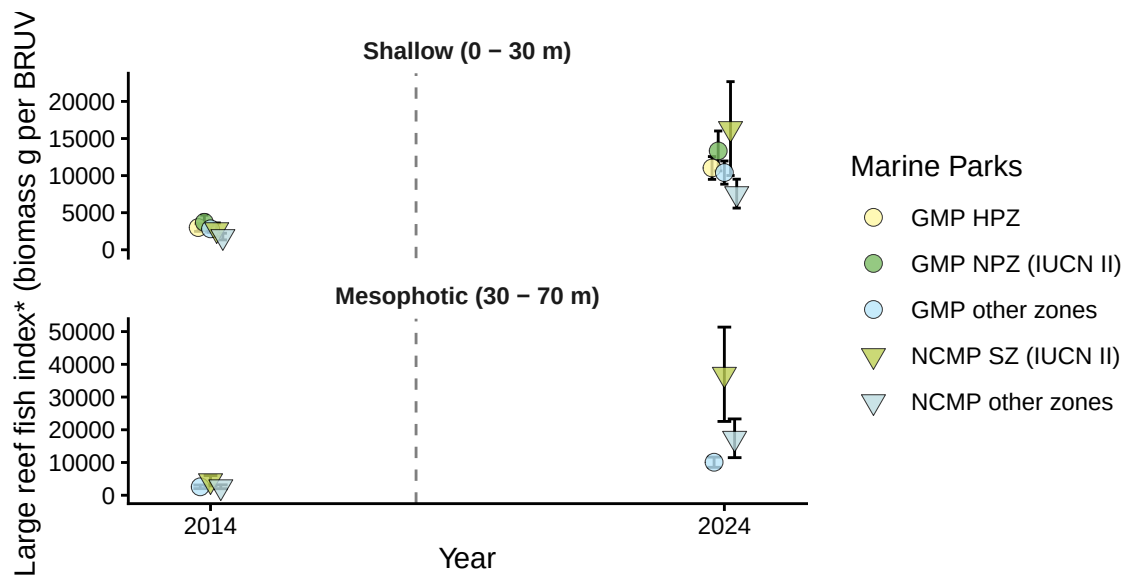


Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

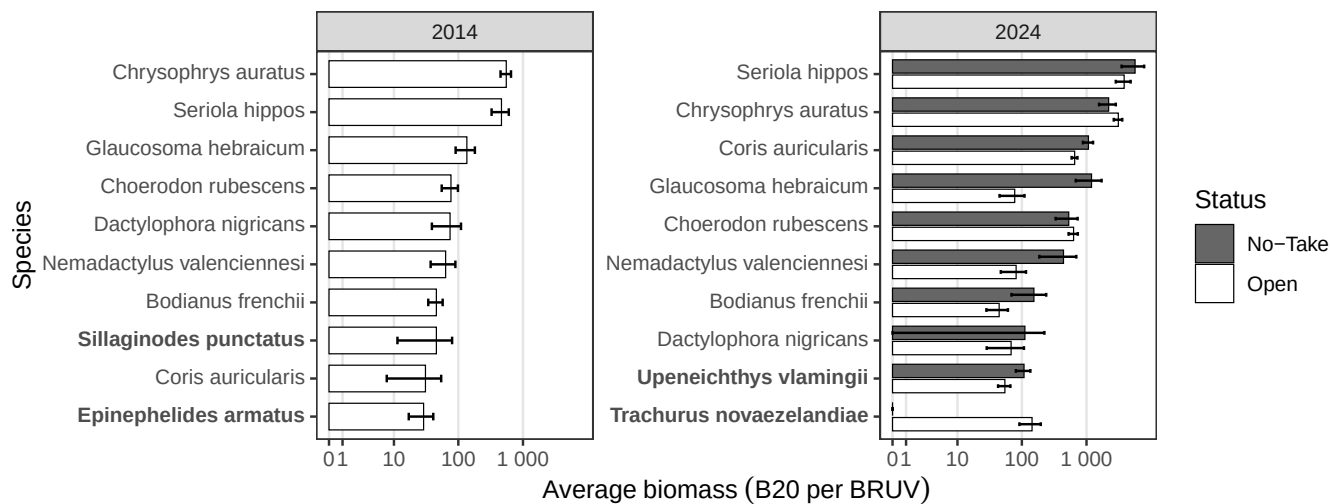


Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species by status (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Large Reef Fish Index*

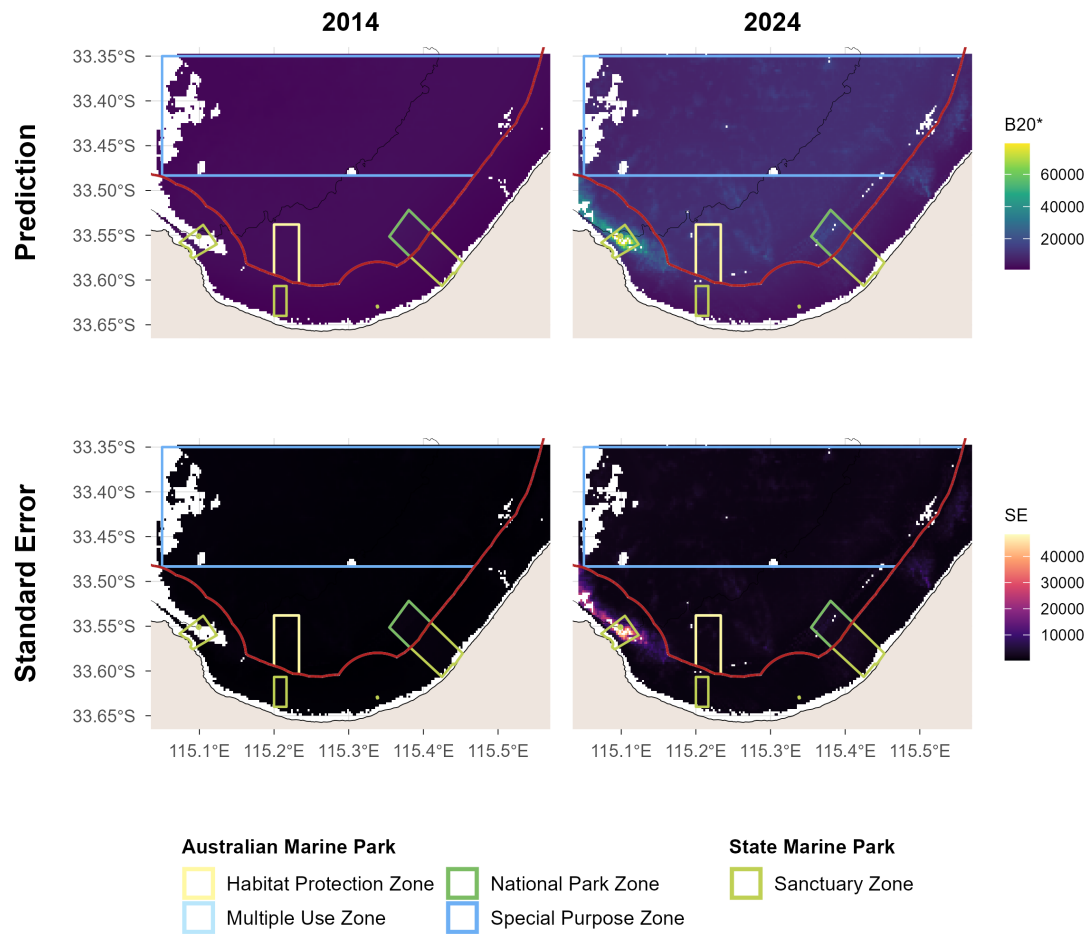


Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

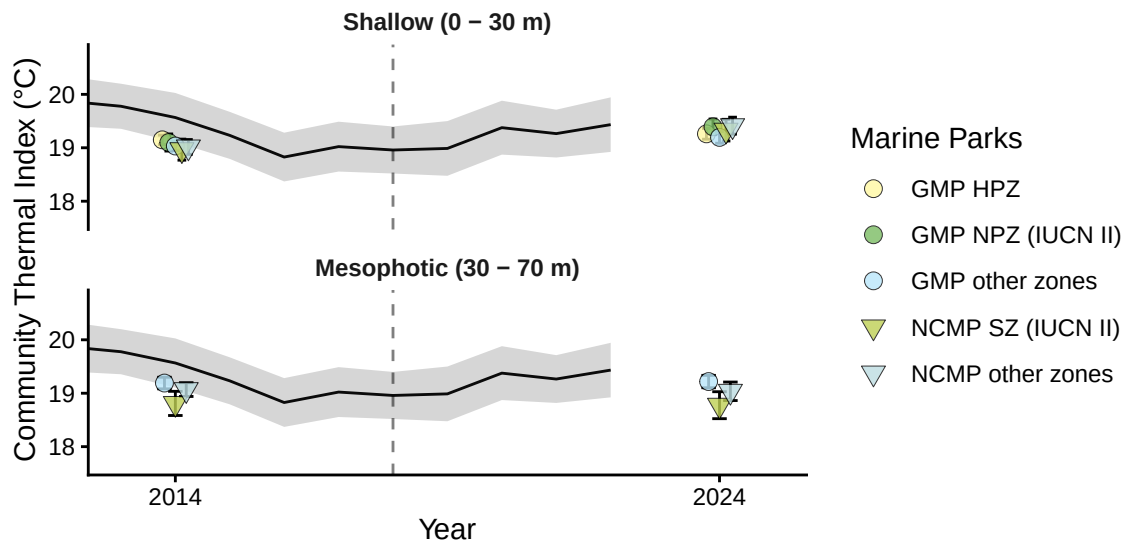


Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

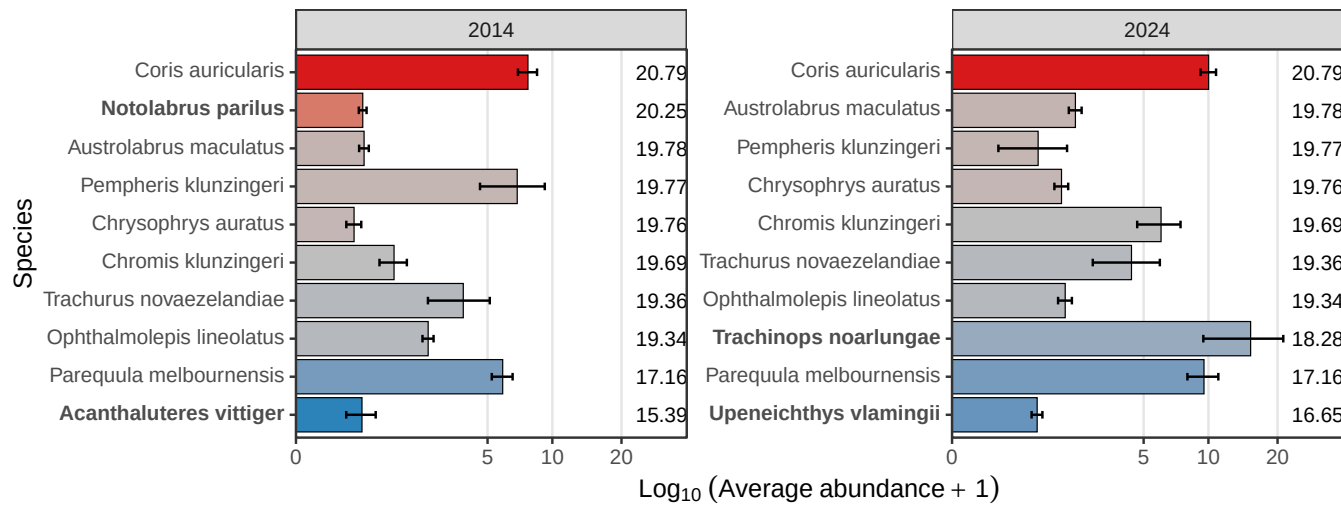


Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Reef fish thermal index

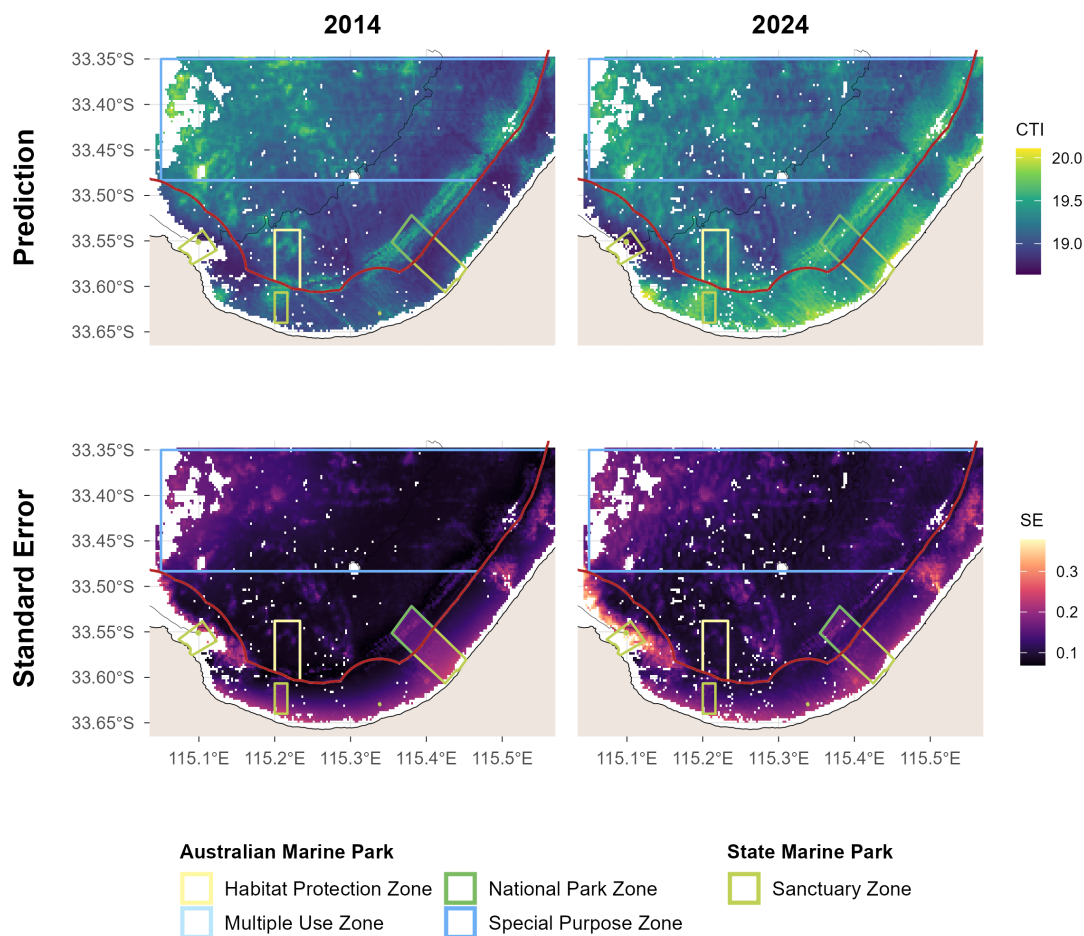


Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.